Venkata Lakshmi Parimala Pasupuleti

Software Engineer · AI Evaluation Frameworks · Agentic Systems · MCP · LangGraph · LLM Evaluation · Observability · Python · Scalable Backend · Generative AI · Responsible AI

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PROFESSIONAL SUMMARY

Software engineer specializing in **automated evaluation frameworks for AI agents** and **agentic system design** - with 8+ years of production engineering depth. Built **AgentProbe**: an open-source **automated evaluation framework** measuring agent performance, accuracy, and safety across 1,240+ test cases - 5-metric eval engine (correctness, groundedness, hallucination, latency, token efficiency), **LLM-as-judge scoring, pass@k harness, and CI-integrated regression gate** blocking quality regressions before merge; observability dashboard tracking per-run metrics, retrieval drift, and latency. Built **Leasel**: an 8-agent LangGraph agentic system with composable agent skills (Triage, Retrieval, Configuration, Reasoning, Validation, Human Escalation) - **responsible AI enforced in code**: LLM reasons, deterministic logic makes every regulated call. **MCP (Model Context Protocol)** architecture in AgentProbe: any agent integrates via REST without code changes. At Entrata: owned enterprise-scale Python systems serving **100K+ MAU at 99.5% uptime** across 120+ Agile sprints, full SDLC ownership end-to-end. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · LangGraph · RAG · Responsible AI

- Built a **LangGraph agentic workflow** (parse → plan → retrieve → validate → conditional routing) applying Chain-of-Thought reasoning with **RAG-based grounding** over FAA Part 107 regulations; enforced **responsible AI safety rules** - every compliance decision made by deterministic logic, not the LLM, blocking non-compliant actions regardless of model confidence.

George Mason University
Aug 2024 – May 2026

Research Associate / Graduate Teaching Assistant — Python · ML Pipelines · Observability · Curriculum

- Built multi-source Python prediction pipelines (190+ countries) — **R^2 = 0.977**; applied Isolation Forest for automated data integrity checks; designed Tableau and ggplot2 **observability dashboards** improving research comprehension 40%. Delivered Python, OOP, and database design workshops to **600+ students**.

Entrata — Enterprise PropTech SaaS
Sep 2018 – Aug 2024 · Remote

Senior Software Engineer — Python · Full SDLC Ownership · Observability · CI/CD · Scalable Backend · AWS

- **Full SDLC ownership of 20+ production modules** (Python, PHP, JavaScript, PostgreSQL, RabbitMQ, AWS S3) serving **100K+ MAU at 99.5% uptime** across 120+ Agile sprints - design documentation, implementation, testing coverage, deployment, and post-launch maintenance for every release cycle.
- Built **internal telemetry and observability tooling** - debug dashboards, structured logging, and performance monitoring reducing debugging time 40% and keeping open bugs below 10 for 6+ months; implemented **CI/CD pipelines** (GitHub Actions) with automated testing gates; **optimized 200+ SQL queries** (30% performance gain); led Docker microservices redesign (50% scalability improvement, 30% deployment reduction).
- **Integrated AI/ML pipeline** (Python, PyTorch, OpenAI API) replacing manual Excel workflows on 1.6M+ federal records; **300+ peer code reviews, mentored 5+ engineers**; conducted RCA on production incidents; led PHI-compliant AWS S3 migration (100% compliance, zero security incidents).

Entrata India (Db Xento Systems)
Jan 2017 – Sep 2018 · Pune

Associate / Trainee Software Engineer — Python · Backend · REST APIs · Data Pipelines

- Built Python/PHP backend modules and REST API integrations for property management and compliance; designed SQL import pipelines for 1.6M federal housing records; led AWS S3 migration; contributed to code reviews and defect analysis under senior engineer guidance.

PROJECTS

AgentProbe: Automated Evaluation Framework for AI Agents — MCP Architecture
AI EVAL · OPEN SOURCE

Python · MCP (Model Context Protocol) · LangGraph · FastAPI · ChromaDB · OpenAI / Anthropic / Groq / Ollama · React · CI/CD (GitHub Actions) — github.com/venkatalakshmiparimala/AgentProbe

- **Automated evaluation framework** measuring AI agent performance, accuracy, and safety: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **LLM-as-judge scoring**, pass@k harness with multi-trial sampling, and **CI-integrated regression gate** blocking quality regressions before merge; **MCP architecture** - any agent integrates via REST/JSON without code changes; React **observability dashboard** with per-run metrics, retrieval drift, and latency tracking; multi-provider generative AI routing (OpenAI, Anthropic, Groq, Ollama). Validated across 1,240+ test cases: pass@1 81.7%, pass@3 88.3%.

Leasel: 8-Agent Agentic System — Composable Skills · HITL · Responsible AI
AGENTIC · LANGGRAPH

Python · LangGraph (8-agent StateGraph) · FastAPI · Streamlit · RAG · ChromaDB · HUD API · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **8-node LangGraph agentic system with composable agent skills** (Triage, Retrieval, Configuration, Historical Incident, Structured Data, Reasoning, Validation, Human Escalation) applying ReAct-style reasoning to automate complex compliance workflows; **HITL escalation** when agent confidence is insufficient; live HUD regulatory API + RAG over compliance documents; **safety enforced in code**: LLM reasons, deterministic logic makes every regulated call; CI regression gate blocks regressions automatically. Results: 93.6% precision, 100% citation correctness, 0% unsupported-claim rate on 1,240 compliance documents.

AI on FHIR: NLP-Powered Web Application & Scalable Backend
WEB APP · BACKEND

Python · Flask · React · BERT · LLMs · FHIR REST API · Docker · spaCy · Embeddings — github.com/venkatalakshmiparimala/AI_on_FHIR

- Flask + React **web application** with **scalable backend** integrating LLM-powered NLP with FHIR REST API - translating natural-language queries (EN/ES) into structured API calls using BERT and embeddings; ~60% reduction in manual retrieval time; HIPAA-compliant RBAC; Docker containerized. Demonstrates full-stack AI/ML platform deployment for cloud-scale data access.

EDUCATION

M.S. Data Analytics Engineering — George Mason University
May 2026 · GPA 3.87

AI systems, ML pipelines, statistical modeling, data engineering, responsible AI, observability

B.Tech Computer Science & Engineering — JNTU Anantapur
2013 – 2017

TECHNICAL SKILLS

Engineering & Backend: Python (expert), PHP, SQL (expert — 200+ production optimizations, schema design, normalization, execution plan tuning, indexing, partitioning, triggers, audit tables), PL/SQL, JavaScript, TypeScript, R, HTML/CSS; REST APIs, Scalable Backend Services, Microservices, Docker, CI/CD (GitHub Actions), Automated Testing, Observability, Monitoring & Alerting, Logging Frameworks, AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Git, Linux

AI & LLM Evaluation: MCP (Model Context Protocol), LangGraph, LangChain, Agentic Workflows, Multi-Agent Orchestration, HITL, ReAct, Automated Evaluation Frameworks, LLM-as-Judge, Pass@k Harness, CI Regression Gate, Agent Safety & Accuracy Testing, RAG Pipelines, Generative AI, LLM Integration (OpenAI, Anthropic, Groq, Ollama), Prompt Engineering, Responsible AI, Guardrails, ChromaDB, FastAPI, NLP, BERT

ML & Analytics: PyTorch, scikit-learn, XGBoost, Random Forest, Isolation Forest, Feature Engineering, Anomaly Detection, Statistical Modeling, Tableau, Power BI, ggplot2, Agile/Scrum, Planning Poker; 300+ code reviews, 5+ engineers mentored